

E-COMMERCE LOGISTICS SYSTEM NETWORK DESIGN (FINAL)

INTRODUCTION

You are tasked with designing a network for an E-Commerce Logistics System connecting six operational centers. The network must support internal communication, web access, automated IP assignment, and controlled routing.

CENTERS & DEVICE COUNT

MOH (Main Operations Hub) – 220 devices

WHC (Warehouse Center) – 160 devices

DDU (Delivery Dispatch Unit) – 130 devices

CSC (Customer Support Center) – 140 devices

DPU (Data Processing Unit) – 90 devices

RPP (Regional Pickup Point) – 60 devices

DEVICE REQUIREMENTS (PER CENTER)

1 router

1 internal switch

2 PCs

Additional Servers by Location

MOH: DNS server + Email server

CSC: Web server (www.orders.local) + Email server + Dedicated DHCP server

WHC: Router-based DHCP

DDU: Router-based DHCP

MOH, RPP, DPU: Static IP only (no DHCP)

TOPOLOGY REQUIREMENTS

Each center has 1 router and 1 internal switch.

MOH and CSC are connected via a common shared switch.

WHC connects to MOH via a direct link.

WHC connects to DDU via a direct link.

DDU connects to RPP via a direct link.

CSC connects to DPU via a direct link.

WHC and CSC have a backup direct link (used only for floating static route).

No other direct connections between CSC and WHC beyond the backup link.

DDU acts as the intermediate node between WHC and RPP.

SERVICES SETUP

Web server: Only in CSC, domain `www.orders.local`

DNS server: Only in MOH, resolves `www.orders.local`

Email servers: Only in MOH and CSC

All other units must send emails through CSC (CSC acts as mail relay)

DHCP CONFIGURATION

WHC: Router-based DHCP

DDU: Router-based DHCP

CSC: Dedicated DHCP server (not on router)

MOH: Static addressing

RPP: Static addressing

DPU: Static addressing

IP ADDRESSING (VLSM)

Generate Base Network Address

Use the following rules based on your group size:

2-member group: Use the last 2 digits of the second member's ID

3-member group: Use the middle 2 digits of the third member's ID

4-member group: Use the sum of the last 3 digits of the fourth member's ID

The derived value forms the first two octets. The remaining octets are 0.0 with a /16 prefix.

Example (4-member group):

Fourth member ID: 20219876

Last 3 digits: 876

Sum = $8 + 7 + 6 = 21$

Base Network: 21.21.0.0/16

ROUTING CONFIGURATION

Static Route Types Required

Next-hop static route: From MOH to DDU, using WHC as the next-hop IP address.

Exit-interface static route: From RPP to MOH, using DDU's exit interface.

Recursive static route: From DPU to RPP, with next-hop CSC. CSC must have a recursive route to RPP via MOH → WHC → DDU.

Floating static route: From WHC to CSC via the backup direct link, with a higher administrative distance than the primary dynamic route.

Default static route: On RPP, pointing to DDU.

Dynamic Routing (RIPv2)

Implement RIPv2 on the following routers: MOH, WHC, DDU, CSC.

Do NOT enable RIPv2 on DPU or RPP.

Mixed Routing on MOH, MOH must use both static routes (to reach all networks manually) and RIPv2 (to exchange routes with WHC, DDU, and CSC).

Route Redistribution

On MOH, redistribute all static routes into the RIPv2 domain.

Disallowed Route Types

Summary static routing is NOT allowed.

Fully specified static routing (using both exit interface and next-hop IP) is NOT allowed.

SPECIAL CONDITIONS (FAILURE SCENARIOS)

If the primary link between WHC and MOH fails, WHC must reach CSC via the backup direct link using the floating static route.

If the primary dynamic path from WHC to CSC fails, the floating static route must automatically become active.

If the direct link between DPU and CSC fails, DPU must still reach RPP using the recursive static route (DPU → CSC → MOH → WHC → DDU → RPP). This requires CSC to have a recursive route to RPP.

Only RPP may have a default route (0.0.0.0/0). No other router may have a default route.

DELIVERABLES

Work distribution among group members

Packet Tracer file (.pkt)

Topology diagram with subnet labels

VLSM tree diagram

IP address table (all devices and interfaces)

Configuration commands for all routers